

Edonis Elshani

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EDUCATION

Technical University of Munich (TUM)

Oct. 2023 – Sep. 2026

M.Sc Electrical Engineering and Information Technology

Munich, GER

- Focus: Robotics, Optimization & Decision Making, Autonomous Systems, Machine Learning
- Grade: 1.3 (German scale; 1.0 = best)

The University of Queensland (UQ)

Feb. 2025 – Sep. 2026

M.Eng Engineering Science (Management) | Double Degree

Brisbane, AUS

- Focus: Entrepreneurship, International Business, Power Systems, Electricity Markets
- GPA: 6.75/7.00

Technical University of Munich (TUM)

Oct. 2019 – Jul. 2023

B.Sc Electrical Engineering and Information Technology

Munich, GER

- Grade: 2.2 (German scale; 1.0 = best)

WORK & RESEARCH EXPERIENCE

Harvard University

Feb. 2026 – Aug. 2026

Research Fellow (M.Sc Thesis)

Cambridge, MA, USA

- Developed a structure-preserving trajectory-optimization framework in **Python, SPOT, and MOSEK**, recovering feasible trajectory with a **0.0002% certified optimality gap**.
- Designed a semidefinite programming (SDP) based model predictive controller (MPC) for constrained Acrobot swing-up and stabilization, successfully **stabilizing 94 of 100** sampled initial states.

BMW Group

Sep. 2024 – Feb. 2025

Systems-Engineer Intern (Autonomous Driving Level 3)

Munich, GER

- Developed and implemented a **MATLAB/Simulink** fault-response model covering **60+ failure modes** to simulate degraded operation of Level 3 automated-driving functions.
- Derived sensor requirements and evaluated safety-critical decision strategies aligned with **ISO 26262**, while developing vehicle-test analysis tools and an HMI concept for highly automated driving.

Knorr-Bremse AG

Nov. 2023 – Aug. 2024

Engineering Working Student (Sensors & Integration)

Munich, GER

- Developed and implemented automated test setups in LabVIEW for connector and sensor validation of a globally deployed railway brake-control product with **more than 100,000 units across 350+ projects worldwide**

Munich Institute of Robotics and Machine Intelligence

Feb. 2023 – Jul. 2023

Research Fellow (B.Sc Thesis)

Munich, GER

- Designed and implemented an automated LabVIEW test stand and collected and analyzed data from 720 physical human-robot collision experiments to characterize injury mechanisms and safety risks.

Expleo Group

May. 2022 – Oct. 2022

Systems-Engineering Intern

Munich, GER

- Implemented automated ECU software-validation workflows on hardware-in-the-loop (HiL) systems and evaluated vehicle- and system-level test results

PUBLICATIONS

Towards Unconstrained Collision Injury Protection Data Sets: Initial Surrogate Experiments for the Human Hand

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024

*Robin J. Kirschner, Jinyu Yang, **Edonis Elshani**, Carina M. Micheler, et al.*

HONORS & AWARDS

Heinrich and Lotte Mühlfenzl Foundation Scholarship (*Technical University of Munich*)

Mar. 2026

Dean's Commendation for Academic Excellence (*The University of Queensland*) – Semester 2

Mar. 2026

Dean's Commendation for Academic Excellence (*The University of Queensland*) – Semester 1

Aug. 2025

SKILLS

Technologies: C++, MATLAB/Simulink, Python, Julia, LabVIEW, ROS2, MOSEK, SPOT, LaTeX

Tools: Git, Linux, Slurm, Docker, CANalyzer, PTC Creo (CAD)

Languages: English (C1/fluent), German (native), Albanian (intermediate), Spanish (intermediate)